F18CE2 - MV Calculus and Real Analysis B: Solutions 5

1 (a) f is continuous at a if the limit $\lim_{x\to a} f(x)$ exists, and $\lim_{x\to a} f(x) = f(a)$.

(b) f is continuous at a if for every $\epsilon > 0$ there exists $\delta > 0$ such that

$$|f(x) - f(a)| < \epsilon, \quad \text{for all } x \in A \text{ with } |x - a| < \delta.$$

2 By definition,

$$\begin{aligned}
\lim_{x\to 0^+} f(x) &= \lim_{x\to 0^+} (3x^2 + 1) = 1, \\
\lim_{x\to 0^-} f(x) &= \lim_{x\to 0^-} (x + 2) = 2, \\
\lim_{x\to 0^-} f(x) &\neq \lim_{x\to 0^+} f(x).
\end{aligned}$$

Hence, $\lim_{x\to 0} f(x)$ does not exist so f is not continuous at 0.

3 By definition,

$$\begin{aligned}
\lim_{x\to 1^+} f(x) &= \lim_{x\to 1^+} (x^2 + 3) = 4, \\
\lim_{x\to 1^-} f(x) &= \lim_{x\to 1^-} (x^3 + 2) = 3.
\end{aligned}$$

Hence, $\lim_{x\to 1} f(x)$ does not exist so f is not continuous at 1.

4 f is continuous at all points of $\mathbb{R}$ except possibly -1 and 1.

For continuity at these points we require the left and right limits to be equal:

$$\lim_{x\to -1^-} f(x) = \lim_{x\to -1^+} f(x) \qquad \text{and} \qquad \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x).$$

Evaluating these limits gives:

$$\lim_{x\to -1^-} f(x) = \lim_{x\to -1^-} (cx + d) = -c + d\,, \quad \lim_{x\to -1^+} f(x) = \lim_{x\to -1^+} (ax^2 + bx) = a - b,$$

so that $a - b = d - c$. Also,

$$\lim_{x\to 1^-} f(x) = a + b\,, \quad \lim_{x\to -1^+} f(x) = c + d,$$

so that $a + b = c + d$. Hence $a = d$ and $b = c$.

5 Let $\epsilon > 0$. If $x \neq 0$, then

$$|f(x) - f(0)| = \left| x \sin \frac{1}{x} \right| \leqslant |x|\,,$$

since $f(0) = 0$.

Set $\delta = \epsilon$. Then

$$|f(x) - f(0)| \leqslant |x| < \epsilon \quad \text{if } 0 < |x| < \delta,$$

so f is continuous at 0.

6 We find a sequence (a_n) which converges to 0 with $f(a_n) = 0$ for all n. Since $\sin y = 0$ if $y = 2n\pi$, we let

$$a_n = \frac{1}{2n\pi}.$$

Then $f(a_n) = 0$ for all n and so

$$\lim_{n\to\infty} a_n = 0\,, \quad \lim_{n\to\infty} f(a_n) = 0 \neq f(0).$$

7 We find a sequence (a_n) which converges to 0 with

$$\lim_{n\to\infty} f(a_n) = 1 \neq f(0) = 0.$$

To do this we choose a_n such that $\cos(1/a_n) = 1$ for all $n \geqslant 1$. Putting

$$a_n = \frac{1}{2n\pi} \implies f(a_n) = a_n + 1 \implies \lim_{n\to\infty} a_n = 0\,, \quad \lim_{n\to\infty} f(a_n) = 1 \neq f(0).$$

8 Let $\epsilon > 0$. We have that $|g(0)| \leqslant |f(0)| = 0$ so that $g(0) = 0$. Now, for all x,

$$|g(x) - g(0)| \leqslant |f(x)|.$$

Since f is continuous at 0 and $f(0) = 0$, there exists $\delta > 0$ such that

$$|f(x)| < \epsilon \quad \text{for all } x \text{ with } |x| < \delta$$

Hence, if $|x| < \delta$,

$$|g(x) - g(0)| \leqslant |f(x)| < \epsilon.$$

9 Putting $\epsilon = f(a)/2 > 0$ in the ϵ-δ definition of f is continuous at a, we see that there exists $\delta > 0$ such that

$$|f(x) - f(a)| < \frac{1}{2}f(a) \quad \text{for all } x \text{ with } |x - a| < \delta.$$

Hence,

$$|x - a| < \delta \implies -\frac{1}{2}f(a) < f(x) - f(a) \implies f(x) > f(a) - \frac{1}{2}f(a) = \frac{1}{2}f(a) > 0.$$

10 Putting $x = y = 0$ in $f(x + y) = f(x) + f(y)$ yields $f(0) = 2f(0)$ so $f(0) = 0$.

Suppose that f is continuous at 0 and let $a \in \mathbb{R}$. We want to show that $\lim_{x\to a} f(x) = f(a)$.

To do this, using the given information about f and the previous result, we will put $x = a+h$, and consider the limit as $h \to 0$. Then, from the given information about f,

$$\lim_{h\to 0} f(a + h) = \lim_{h\to 0} \big(f(a) + f(h)\big) = f(a) + \lim_{h\to 0} f(h) = f(a)$$

(since f is continuous at 0, we have $\lim_{h\to 0} f(h) = f(0) = 0$).

11 Putting $x = y = 0$ in $f(x+y) = f(x)\,f(y)$ yields $f(0) = f(0)^2$, so $f(0) = 1$ since $f(0) \neq 0$.

Suppose that f is continuous at 0 and let $a \in \mathbb{R}$. We want to show that $\lim_{x\to a} f(x) = f(a)$.

As in the previous question, we will put $x = a + h$, and consider the limit as $h \to 0$. Then, from the given information about f,

$$\lim_{h\to 0} f(a + h) = \lim_{h\to 0} \big(f(a)f(h)\big) = f(a) \lim_{h\to 0} f(h) = f(a)$$

(since f is continuous at 0, we have $\lim_{h\to 0} f(h) = f(0) = 1$).

12 The Intermediate Value theorem: Let f be continuous on $[a, b]$, and suppose that γ satisfies

$$f(a) \leqslant \gamma \leqslant f(b) \quad \text{or} \quad f(a) \geqslant \gamma \geqslant f(b).$$

Then there exists $c \in [a, b]$ such $f(c) = \gamma$.

Another way of thinking of this: Let f be continuous on $[a, b]$, and suppose that $f(a)$ and $f(b)$ are non-zero and have opposite signs. Then there exists $c \in (a, b)$ with $f(c) = 0$.

Put $f(x) = x^7 - 4x^4 + x + 1$. Then f is continuous on $[0, 1]$ and $f(0) = 1$ while $f(1) = 1 - 4 + 1 + 1 = -1$. By the IVT, $f(c) = 0$ for some $c \in (0, 1)$.

13 Let $f(x) = xe^x + \sin^2 x - 1$. Then f is continuous on $\mathbb{R}$.

Since $f(0) = -1 < 0$ we have to find b such that $f(b) > 0$.

There are many possible choices, we take $b = \pi$ since

$$f(\pi) = \pi e^\pi - 1 > 0.$$

By the IVT, $f(c) = 0$ for some $c \in (0, \pi)$.

14 Define $g : [0, 1] \to \mathbb{R}$ by $g(x) = f(x) - x$. Then g is continuous on $[0, 1]$ and

$$g(0) = f(0) - 0 > 0, \qquad g(1) = f(1) - 1 < 0.$$

By the IVT, $g(c) = 0$ for some $c \in (0, 1)$, that is, $f(c) = c$.

15 Define $g : [0, 1/2] \to \mathbb{R}$ by $g(x) = f(x + 1/2) - f(x)$. Then g is continuous on $[0, 1/2]$ and, since $f(0) = f(1)$,

$$g(0) = f(1/2) - f(0), \quad g(1/2) = f(1) - f(1/2) = g(1/2) = f(0) - f(1/2) = -g(0).$$

Hence:
if $g(0) \neq 0$ then $g(0)$ and $g(1/2)$ have opposite signs, so by the IVT, $g(c) = 0$ for some $c \in (0, 1)$, that is, $f(c + 1/2) = f(c)$;
if $g(0) = 0$ then $f(1/2) = f(0)$.

16 $f(x) = \dfrac{1}{1 - x}$ is continuous but not bounded on $[0, 1)$.